RAG Engineering Issue Log

GenAI Health Risk & Cost Advisor | V3 | Updated 23 October 2026 | Insurance-policy PDF RAG

Scope: Engineering observations, experiments, code changes and decisions for Phase 1. This log records observed results; proposed future work is marked as such.

Issue 001 - Initial retrieval limitations

Problem: The earlier PDF question-answering flow used TF-IDF retrieval and fixed-size character chunking. Keyword mismatch and chunk boundaries could reduce retrieval quality.

Hypothesis: Sentence-aware chunks and semantic embeddings could preserve context and better match paraphrased insurance questions.

Experiment: Built separate V3 embedding and vector-store services using sentence-transformers/all-MiniLM-L6-v2 and ChromaDB; retained the original answering flow while evaluating retrieval offline.

Result: Semantic retrieval and ChromaDB were implemented as experimental services. Quantitative chunking comparison is recorded in Issue 002.

Status: Superseded as the main V3 retrieval baseline by semantic retrieval; previous versions remain preserved.

Issue 002 - Evidence ranked below less-direct passages

Problem: Direct evidence for the 24-month pre-existing-condition waiting period sometimes ranked below less-direct policy passages. One fixed-chunk query missed the direct evidence in top three.

Hypothesis: Sentence-aware chunks may preserve the meaning of insurance clauses better than fixed-size character chunks.

Experiment: Compared fixed-size (1,000 characters, 200 overlap) with sentence-aware (1,000-character target, approximately 200 overlap) chunking. Same MiniLM model, ChromaDB and six policy questions; evaluated Hit@3 and MRR@3 with expected evidence-page labels.

Result: Fixed-size: Hit@3 0.833, MRR@3 0.611. Sentence-aware: Hit@3 1.000, MRR@3 0.722. Both experimental approaches produced 10 chunks. Direct page-2 evidence moved from rank 3 to rank 2 for two questions; one previously missed question retrieved page 2 at rank 3.

Status: Sentence-aware selected for the working V3 baseline, subject to integration/regression checks. Results are from a small six-question benchmark.

Issue 003 - BM25 and weighted hybrid retrieval

Problem: Equal-weight hybrid retrieval missed direct evidence for a paraphrased question even though semantic retrieval found it.

Hypothesis: BM25 can strengthen exact-term matches but weaken paraphrase matching; increasing semantic weight in Reciprocal Rank Fusion (RRF) may recover evidence.

Experiment: Created tests/bm25_experiment.py; compared MiniLM, BM25Okapi and weighted RRF (50/50, 70/30, 80/20) using 10 sentence-aware chunks and the same six questions.

Result: Semantic: Hit@3 1.000, MRR@3 0.722. BM25: 0.833, 0.750. Hybrid 50/50: 0.833, 0.750. Hybrid 70/30 and 80/20: 1.000, 0.722. No tested hybrid configuration improved aggregate results over semantic-only.

Status: Experiment complete; BM25 and hybrid retained for comparison, not integrated into production-facing V3 retrieval.

Issue 003 - Code change record

File: tests/bm25_experiment.py

Added: BM25Okapi keyword ranking; normalized MiniLM cosine-similarity ranking; weighted RRF combining ranked chunk IDs; six-case JSON benchmark; Hit@3 and MRR@3; per-question diagnostic output.

Efficiency: Reused indexed document representations and embeddings across benchmark questions rather than rebuilding them for every query.

Corrections: Fixed evaluation-loop indentation that temporarily produced zero metrics; changed int() to round() in weight-label construction to prevent inconsistent labels such as 80/19.

Use cases: Semantic retrieval for paraphrases; BM25 for exact policy terminology; hybrid for mixed queries.

Decision: No BM25 or RRF change to services/rag_service.py.

Issue 004 - V3 semantic retrieval integration

Problem: The application-facing services/rag_service.py still used TF-IDF despite the V3 semantic retrieval experiments.

Hypothesis: Integrating sentence-aware chunking, MiniLM and ChromaDB would support evidence-backed insurance-policy answers while preserving Gemini grounding and page citations.

Changes: Updated services/rag_service.py: replaced fixed-size chunking with sentence-aware grouping; replaced TF-IDF retrieval with create_vector_collection(), add_documents() and search_similar_documents(); retained PDF validation, LangChain Document metadata, page-labeled context, Gemini prompt and response parsing. Existing embedding and vector-store service interfaces were reused.

Test: Sample five-page fictional insurance policy: PDF extraction returned 11 chunks, first chunk from page 1. For the waiting-period question, top-three retrieved pages were [4, 2, 1]. End-to-end Gemini response correctly stated 24 continuous months and cited page 2.

Result: Integrated semantic retrieval and grounded answer generation passed the observed smoke tests. Torch macOS redirects, Hugging Face unauthenticated requests and Gemini temperature notices were non-blocking.

Status: Working prototype baseline. Not yet validated for multiuser security, persistent indexing or a larger answer-level benchmark.

Issue 005 - Open integration and evaluation limitations

Problem: The integrated sentence-aware chunker produced 11 chunks versus 10 in the offline benchmark; a new ephemeral collection is created and indexed for each question. The direct page-2 clause is second, behind page 4.

Hypothesis: Implementation differences in chunking/overlap can alter the index and ranking; repeated indexing adds avoidable latency. Page-level ranking does not fully measure answer correctness.

Experiment: Not yet completed. Planned: reconcile chunk counts; run unsupported-answer regression tests; extend evaluation to multiple policies and answer-level correctness; index each uploaded document once.

Result: Current end-to-end example answered correctly despite page 2 being rank 2. No verified improvement from additional reranking experiments yet.

Status: Open; non-blocking for starting Phase 2. Avoid claims of production readiness.

Phase decision and next steps

Decision: Stop expanding retrieval techniques for now. Keep sentence-aware MiniLM + ChromaDB as the V3 baseline; preserve BM25/hybrid experiments as evidence of evaluation and trade-off analysis.

Phase 2 - FastAPI: Add upload/index and question-answer endpoints, typed request/response models, input validation and error handling, reusable per-document indexes, document identifiers and basic document-session isolation. Authentication, durable storage, deployment and load testing remain later work.

Interpretation caution: Six questions from one fictional policy, with page-level expected labels, are insufficient to claim broad retrieval superiority. Hit@3 measures whether an expected page appears among three results; MRR@3 rewards its earliest rank. Neither proves answer factuality or production security.

Reference context

Microsoft Azure AI Search RAG overview: https://learn.microsoft.com/azure/search/retrieval-augmented-generation-overview

Azure Databricks retrieval quality guide: https://learn.microsoft.com/en-us/azure/databricks/ai-search/retrieval-quality

Azure RAG retrieval architecture: https://learn.microsoft.com/azure/architecture/ai-ml/guide/rag/rag-information-retrieval